Dataset Name : Tom and Jerry Image Classification Dataset

Dataset Description:

This dataset comprises images extracted from the classic cartoon series "Tom and Jerry." It is organized into four image folders:

- **Tom** – containing images of Tom.
- **Jerry** – containing images of Jerry.
- **tom_jery_0** – a supplementary folder containing images with both characters or ambiguous scenes.
- **tom_jery_1** – another supplementary folder with mixed or challenging scenes.

Additionally, the dataset includes two CSV files:

- **challenges.csv** – highlighting specific challenges or ambiguous cases within the dataset.
- **ground_truth.csv** – providing the accurate labels for the images, which are used for training and evaluation.

Model Selection and Model Training:

For this project, two deep learning architectures were chosen:

- **InceptionV3:** Leveraging the power of transfer learning, this model was adapted by loading pre-trained weights (from a local file) and fine-tuning its top layers. The network's base layers are frozen to retain previously learned features while custom dense layers are added for classification.
- **VGG16:** Similarly, VGG16 was employed using locally stored weights. Its feature extraction layers were frozen, and custom layers were appended to learn the nuances of the dataset.

Results:

After training the models on the "Tom and Jerry" dataset, the evaluation on the validation set yielded the following accuracies:

- **Model 1 (InceptionV3-based):**

```
137/137 ━━━━━━━━━━━━━━━━━━━━ 8s 56ms/step - acc: 0.8257 - loss: 0.4897

W0000 00:00:1740638140.679132      78 graph_launch.cc:671] Fallback to op-by-op mode because m
emset node breaks graph update

35/35 ━━━━━━━━━━━━━━━━━━━━ 1s 32ms/step - acc: 0.7120 - loss: 0.7934
Training accuracy: 0.8231804966926575
Testing accuracy: 0.7068493366241455
```

- **Model 2 (VGG16-based):**

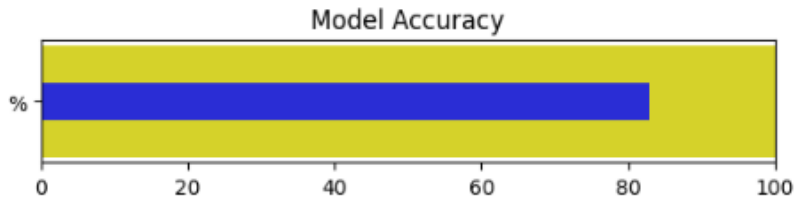
```
137/137 ━━━━━━━━━━━━━━━━━━━━ 6s 41ms/step - acc: 0.9962 - loss: 0.0107

W0000 00:00:1740638148.441377      79 graph_launch.cc:671] Fallback to op-by-op mode beca
use memset node breaks graph update

35/35 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - acc: 0.8320 - loss: 0.7797
Training accuracy: 0.9970340132713318
Testing accuracy: 0.8237442970275879
```

- **Ensemble Model:**
  By averaging the predictions of the two models, the overall accuracy achieved is **82%** on the validation dataset.

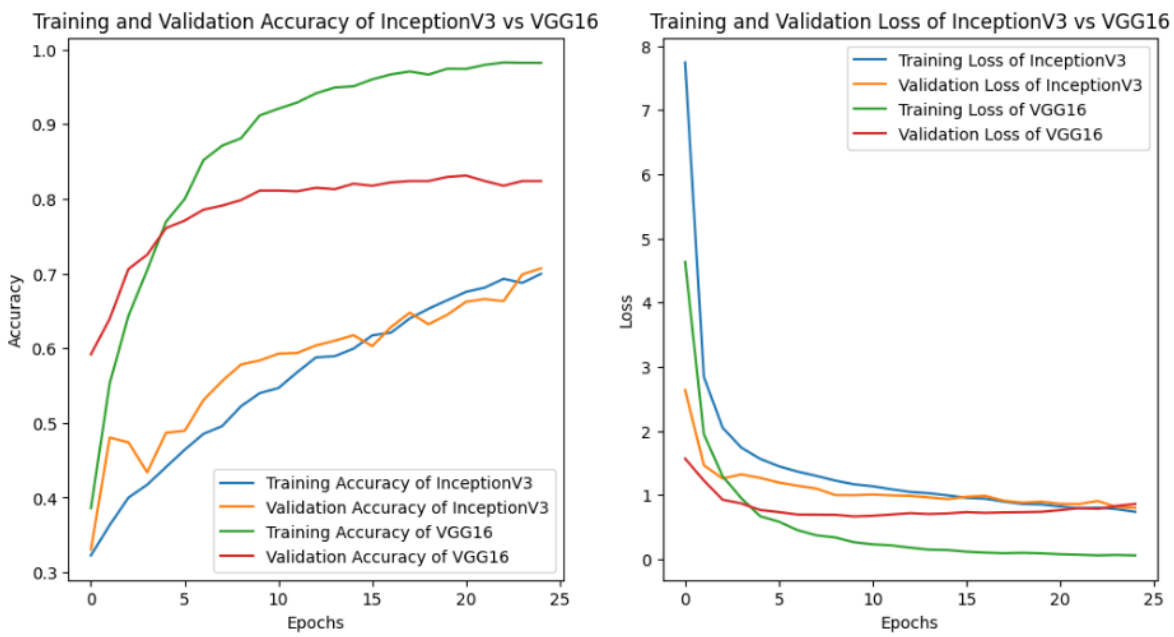
Note: Overall ensemble accuracy is highlighted as the key result, confirming the robustness of the multi-model approach.



Model Performance:

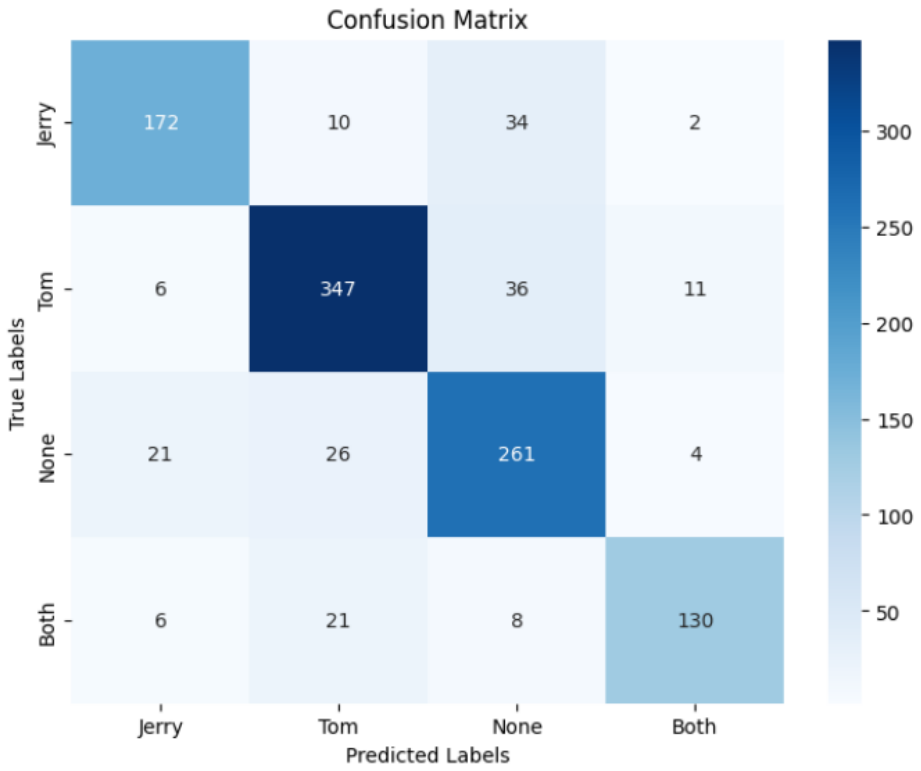
The ensemble approach combines the strengths of both models:

- **Visualization:**



Making Predictions :

Confusion Matrix :



Discussion:

The project achieved an overall accuracy of **82%** on the validation dataset, which indicates a reasonably effective performance for character recognition in the "Tom and Jerry" images. This result is promising given the challenges present in the dataset, such as mixed or ambiguous images in the supplementary folders.